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NTN®

ニカメット®

Magnetic products

CAT. No. 5205/J



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High efficiency and miniaturization to address the is

Customer-specific

Magnetic components for magnetic sensors

ÿ Uses high-precision, miniaturized, and high-magnetic-force isotropic neodymium bonded magnets **ÿ High design flexibility,** allowing integration with surrounding components and handling of complex shapes

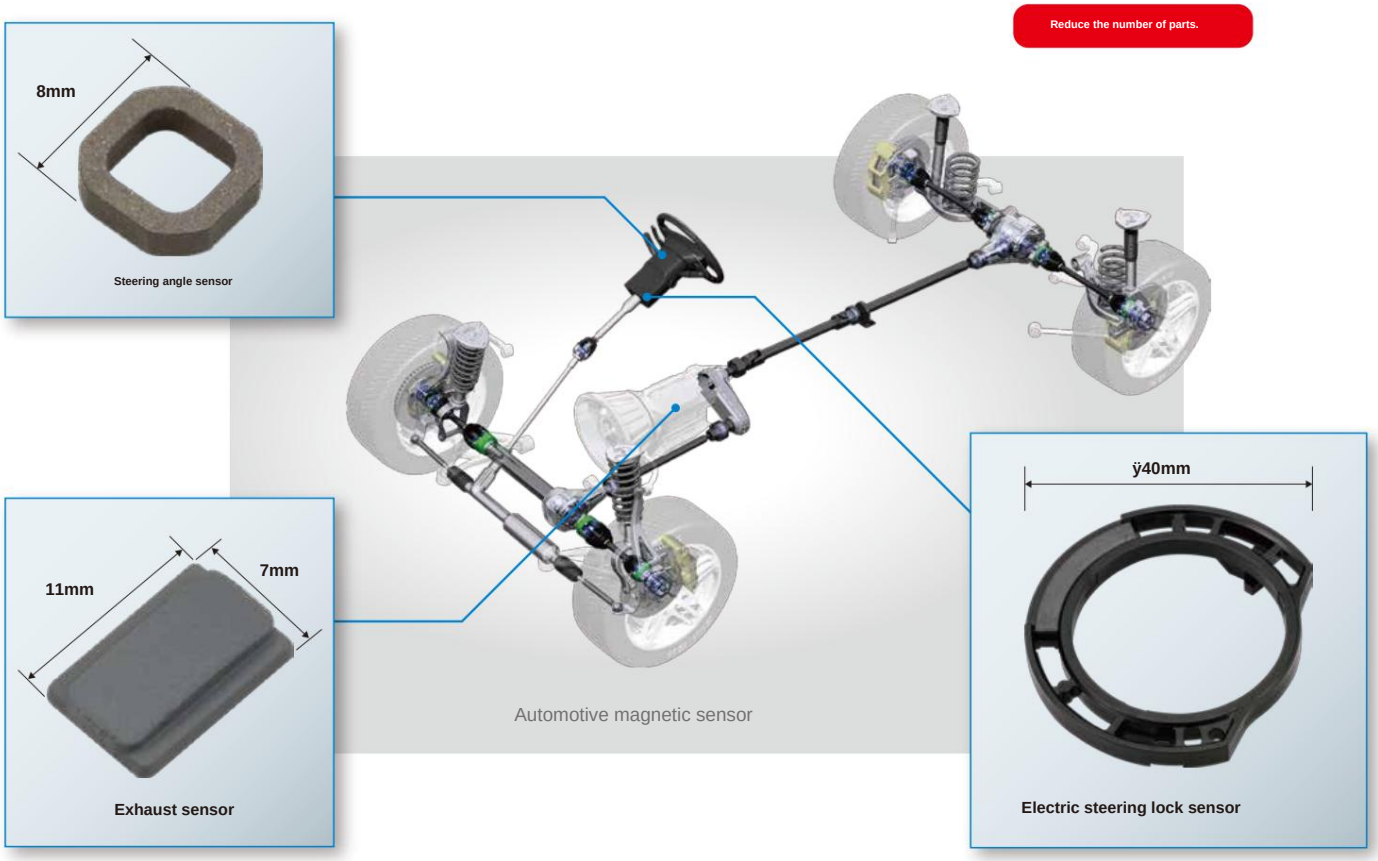
Application examples

ÿ Steering angle sensor/exhaust sensor:

Uses high-strength isotropic neodymium compression bonded magnets* to detect angle and position with high precision.

ÿ Electric steering lock sensor: Isotropic neodymium injection-bonded magnet* is insert-molded into resin parts.

Reduce the number of parts.



ÿ Earthquake sensor:

A unit combining multiple magnets.

Volume reduced by 80% compared to conventional models.



*Magnetic properties are listed on the back cover.

"Nikamet" is a general term for products made from original raw materials manufactured by NTN Advanced Materials Corporation (formerly Nippon Kagaku Yakin), and is a coined word combining "Nikka" (ÿÿ) and "metal".

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NTN's magnetic products

IZ Design and Simulation Support

Choke coil reactor

ÿ Uses a unique amorphous magnetic material that offers **high efficiency, miniaturization,** low loss, and high-frequency compatibility . **Our injection-molded amorphous magnetic material is one of a kind.**

Application examples

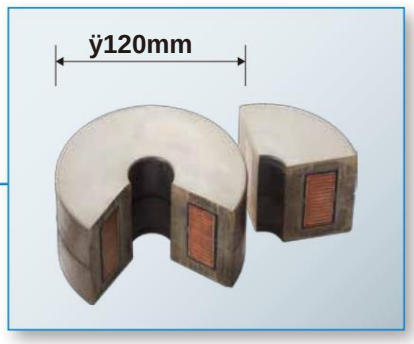


MRI (Magnetic Resonance Imaging)

ÿ Power supply choke coil

By using injection-molded amorphous magnetic material*, heat generation is reduced.

Volume reduced by 80% compared to conventional models.





PHEV•HEV•EV

ÿ Reactor for inverter/converter

By using compressed amorphous magnetic material*, energy loss is reduced.

Energy conservation

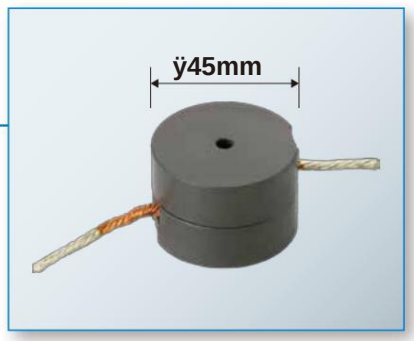




Solar power generation equipment

ÿ Power conditioner reactors use injection-molded amorphous magnetic material to improve power conversion efficiency.

Increase in power generation



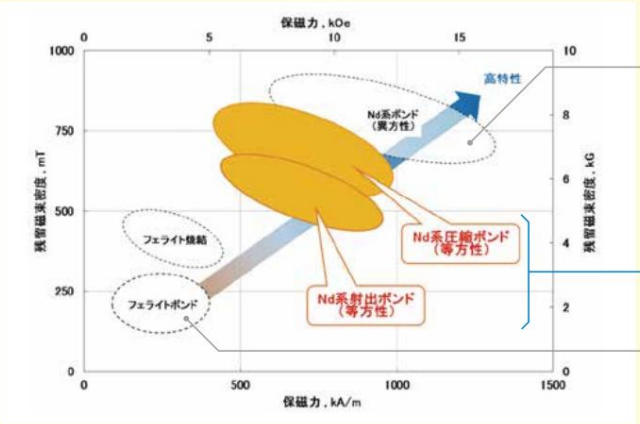
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Material properties

Compression molding offers superior magnetic properties, **while injection molding offers exceptional shape flexibility.**

Magnetic materials for magnetic sensors



High performance, expensive

Isotropic neodymium bonded magnets

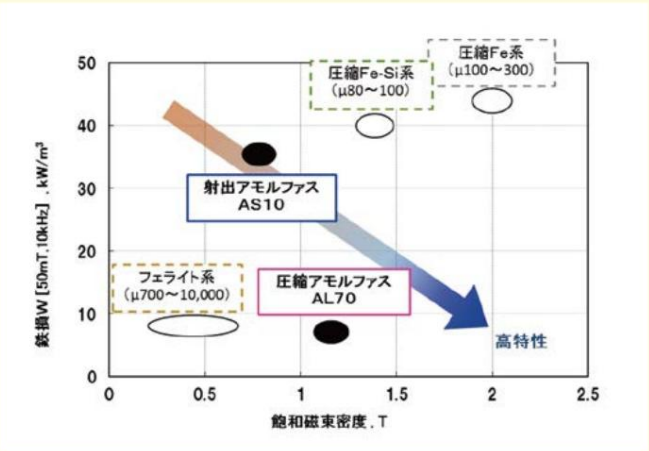
• Higher magnetic properties than ferrite-based bonds • Cheaper than anisotropic neodymium-based bonds

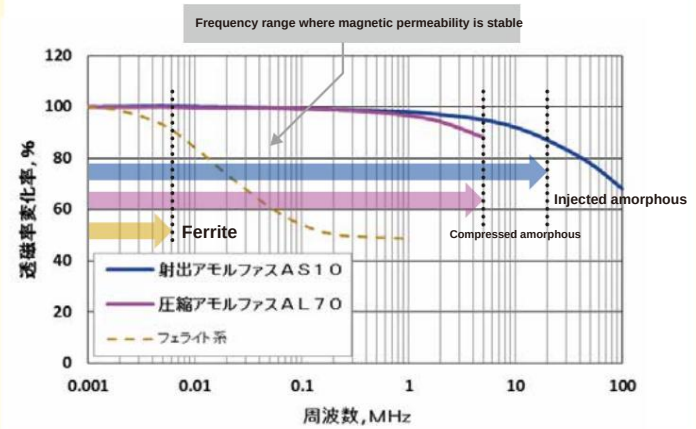
We will propose the most suitable materials according to your specifications.

Low performance, low cost

Materials for choke coil reactors

Compressed amorphous magnetic material AL70	• Highest permeability among compressed amorphous materials • Saturation magnetic flux density approximately twice that of ferrite • Low iron loss comparable to ferrite despite being a compressible material
Injection-molded amorphous magnetic material AS10	• Lower iron loss than compressed Fe-Si systems • Approximately 1.5 times higher saturation magnetic flux density than ferrite • Applicable to high frequencies exceeding 10 MHz





Magnetic Materials Terminology Collection

Anisotropic	A magnet that produces a strong magnetic force only in a specific magnetization direction. Requires special molding
magnets, isotropic	equipment. A magnet that produces the same magnetic force regardless of the magnetization direction, without directional constraints.
magnets, bonded	A magnet made by mixing magnetic material powder into rubber or plastic and molding it. A magnet
magnets,	manufactured by compressing metal powder and firing it at high temperatures. A magnet whose main
sintered magnets, ferrite	component is iron oxide (Fe2O3). Inexpensive and highly corrosion-resistant.
magnets, neodymium	Nd-Fe-B alloy magnet. Extremely strong magnetic force. Amorphous
magnets, amorphous magnetic	alloy magnetic material. Minimal decrease in power-to-magnetic force conversion ability even at high frequencies. The magnetic
materials, residual	force remaining after magnetization of the material. It will be below the saturation magnetic flux density shown below.
magnetic	The reverse magnetic force required to completely remove the magnetic force. An indicator of susceptibility to degradation. The
flux density, coercivity,	maximum magnetic force per unit area that the material can generate. Energy loss that
	occurs when magnetic force is generated in a magnetic material. Released as heat. A material property that indicates the ability
saturation magnetic flux density, iron loss, permeability	to convert electricity into magnetic force, or the ease with which magnetic force is transmitted.

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